

Recurrence and Survival in Patients With Merkel Cell Carcinoma Undergoing Surgery Without Adjuvant Radiation Therapy to the Primary Site

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IMPORTANCE The use of adjuvant radiation therapy (RT) to the primary site in Merkel cell carcinoma (MCC) is not uncommon. However, the need for adjuvant RT to the primary site in patients at low risk for local recurrence is questionable.

OBJECTIVES To examine the occurrence of true local, satellite, in-transit, regional, and distant recurrences in patients undergoing surgery alone without adjuvant RT to the primary site. To establish overall survival (OS), MCC-specific survival (MCCSS), and disease-free survival (DFS) relationships in a cohort of patients with MCC.

DESIGN, SETTING, AND PARTICIPANTS Our University of Michigan Multidisciplinary MCC Program database was used to obtain characteristics and outcome measures for 104 patients (105 primary MCCs) with tumors less than 2 cm in diameter. The majority of patients were treated between July 2006 and November 2012.

MAIN OUTCOMES AND MEASURES Outcome measures included the occurrence of true local, satellite, in-transit, regional, and distant recurrences. End points included OS, MCCSS, and DFS.

RESULTS Overall, information for 55 men and 49 women with 105 primary MCCs was obtained; 19 patients developed recurrent disease, and the mean time to first recurrence was 10.7 months. True local recurrence occurred in 1 patient with concurrent in-transit recurrence. Satellite recurrence occurred in 1 patient with concurrent regional recurrence. Four additional patients developed in-transit metastases. Thirteen patients had a regional recurrence component, 4 patients had distant metastases, and 6 patients developed subsequent regional and/or distant recurrences. Stratified by initial pathologic stage, the OS and MCCSS at 48 months were estimated to be 85.0% (95% CI, 71.8%-92.3%) and 94.4% (95% CI, 83.4%-98.2%) for patients with stage 1A/B disease and 63.2% (95% CI, 36.6%-81.1%) and 78.1% (95% CI, 50.0%-91.5%) for patients with stage 3A disease. The OS and MCCSS at 24 months for patients with stage 3B disease were both 50.0% (95% CI, 5.8%-84.5%).

CONCLUSIONS AND RELEVANCE In selected MCC patients with primary tumors less than 2 cm in diameter treated with surgery alone without adjuvant RT to the primary site, we found a low occurrence of true local recurrences and satellite recurrences. This relatively low rate of local recurrence questions the need for adjuvant RT to the primary tumor site in patients with small low-risk lesions.

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Merkel cell carcinoma (MCC) is a potentially aggressive cutaneous malignancy with a considerable risk of local recurrence, regional and distant metastasis, and death. While relatively uncommon, the incidence of MCC tripled in the last 2 decades of the twentieth century and continues to increase.¹ The majority of patients (>70%) present with disease that is clinically localized to the primary site (2010 American Joint Committee on Cancer [AJCC] stage 1 or 2); however, high-level evidence regarding optimal management of the primary tumor site is lacking.²⁻⁴ Local recurrence rates vary widely and have been reported between 4% and 53%.³⁻⁷ Precise local recurrence rates are difficult to ascertain as regional failures are frequently included as locoregional recurrence.

An optimal strategy for management of the primary tumor site has yet to be determined, and the recommendations in the literature are variable. Given the limitations in the available literature, our primary objective was to examine the occurrence of local, satellite, in-transit, and regional recurrence, as well as distant metastasis in a contemporary cohort of patients with MCC undergoing wide local excision of the primary tumor site without the use of adjuvant radiation therapy (RT) to the primary site. A secondary objective was to establish overall, Merkel cell carcinoma-specific survival (MCCSS), and disease-free survival (DFS) relationships within this cohort.

Methods

The institutional review board at the University of Michigan approved this study. Our prospectively maintained MCC database was queried for all patients who did not receive adjuvant RT to the primary site following wide local excision of a primary MCC at the University of Michigan. One hundred twelve patients were identified. The 8 patients with primary tumors 2 cm or larger were excluded because adjuvant RT to the primary site was recommended but the patients declined, or complicated postoperative wound healing prevented administration. In total, 104 patients with 105 primary MCCs less than 2 cm were included. The majority of the tumors (n = 102) were evaluated and treated between July 2006 (the opening of the MCC program) and November 2012. Three additional tumors treated at the University of Michigan between 2001 and 2005 were included because these patients continued their follow-up in our MCC clinic. Patients were presented to our Multidisciplinary MCC Tumor Board for consensus regarding the initial treatment plan. All patients underwent wide local excision of the primary tumor. Surgical margins of generally 1 to 2 cm were excised based on the clinical presentation and location of the tumor, allowing for, whenever possible, primary closure without a skin graft. Tumors with a positive surgical margin were re-excised. None of the patients received adjuvant RT to the primary site(s). Regional staging via sentinel lymph node (SLN) biopsy at the time of the wide local excision was recommended for all clinically node-negative patients at presentation. Patients with palpable regional nodal disease at presentation underwent completion

Key Points

Question What is the occurrence of true local, satellite, in-transit, regional, and distant recurrences in patients with Merkel cell carcinoma undergoing surgery without adjuvant radiation therapy to the primary site?

Findings In this cohort study of 104 patients with 105 primary tumors less than 2 cm in diameter, true local with concurrent in-transit recurrence occurred in 1 patient, satellite with concurrent regional recurrence occurred in 1 patient, 4 additional patients developed in-transit recurrences, 13 patients had regional recurrences, and 4 patients had distant metastases.

Meaning Low occurrence of true local and satellite recurrences questions the need for adjuvant RT to the primary site in selected patients with small tumors.

lymph node dissection at the time of the wide local excision following tissue confirmation of nodal metastasis by fine needle aspirate or excision. Adjuvant RT to the regional basin following completion lymph node dissection was recommended for individuals at high risk for regional recurrence (ie, ≥ 2 lymph nodes involved, large nodal tumor burden, and/or extracapsular extension). Regional RT consisted of a cumulative dose of at least 50.4 Gy (to convert milligray to rad, multiply by 0.1) to the treated area. None of the patients received chemotherapy as part of their initial treatment.

Clinical characteristics included patient sex, age, presence or absence of immunosuppression (medication- or disease-induced), site of the primary MCC (categorized as head/neck, upper extremity, lower extremity including buttock, and trunk), and the clinical diameter of the primary tumor in centimeters (categorized as <1 cm and 1-2 cm). Histologic characteristics of the primary tumor included tumor thickness (measured in millimeters from the granular layer of the epidermis to the deepest extent of tumor invasion [ie, Breslow thickness]), presence or absence of angiolymphatic invasion, number of mitoses per millimeter squared, growth pattern, presence or absence of ulceration, and nodal disease status (positive or negative SLN or clinical lymph node metastasis at presentation). All patients were staged according to the 2010 AJCC staging for Merkel cell carcinoma.⁸ Histopathologic evaluation of all primary lesions and SLN biopsies was performed by board-certified dermatopathologists in the MCC program at the University of Michigan.

Outcome measures included the occurrence of true local recurrence (recurrent tumor arising within, or directly adjacent to, the primary excision scar), satellite recurrence (cutaneous metastases within 2 cm of the primary excision scar), in-transit recurrence (cutaneous metastases >2 cm from the primary excision scar but distal to the regional draining lymph node basin), regional recurrence (palpable or image-based recurrence within the regional draining lymph node basin), distant metastases, and death. End points included overall survival (OS), MCCSS, and DFS. Follow-up time for the end points were calculated from the date of diagnosis until death (ie, OS), until death due to Merkel cell carcinoma (ie, MCCSS), or until first occurrence of local, satellite, in-transit, regional recurrence or distant metastasis. Patients not experiencing the

events of interest were censored on their last known clinical follow-up date. The product-limit method of Kaplan and Meier was used to estimate event-free probabilities, and Cox proportional hazard regression models were used to explore possible associations with the patient and tumor characteristics. Multivariable models were constructed for OS and DFS. All clinical and histopathologic characteristics as previously described were presented to the models; however, because of the strong association between tumor diameter and depth, these were not offered simultaneously to the multivariate models. The best multivariate models were constructed based upon consideration of statistical significance and the Akaike Information Criterion. For all statistical tests, *P* values at or below .05 were considered significant.

Results

Patient and Tumor Data

One hundred four patients (55 males, 49 females) with 105 primary MCCs were identified; 64 tumors were less than 1 cm in clinical diameter, and 41 tumors were 1 to 2 cm. The mean histopathologic tumor thickness was 3.9 mm, and the mean mitotic rate was 26.0 mitoses/mm². Angiolymphatic invasion was identified in 17 tumors and absent in 80. The mean surgical margin was 1.4 cm for tumors less than 1 cm and 1.5 cm for tumors 1 to 2 cm in diameter. At presentation, 100 patients with 101 tumors were clinically node negative. Sentinel lymph node biopsy was recommended for each of these patients. For the 24 SLN-positive tumors, completion lymph node dissection was performed in 21 cases. Four patients presented with palpable lymphadenopathy (stage 3B disease). Each of these patients underwent a completion lymph node dissection following tissue confirmation of nodal metastases. The initial AJCC pathologic staging for this cohort was 69 patients (65.7%) with stage 1A disease; 8 (7.6%), stage 1B; 24 (22.9%), stage 3A; and 4 (3.8%), stage 3B. Additional patient and tumor data are listed in eTables 1 and 2 in the [Supplement](#), respectively.

None of the 105 primary tumor sites were treated with adjuvant RT. However, 8 patients received adjuvant RT to the regional lymph node basin as part of the initial treatment plan. Two patients with a positive SLN biopsy declined completion lymph node dissection and were treated with primary RT to the nodal basin. One patient with a positive SLN biopsy declined further treatment and MCC recurred in the regional nodal basin 6 months later. Completion lymph node dissection was performed for this patient at the time of recurrence.

Follow-up

The median follow-up for this cohort of patients was 34.3 months (95% CI, 29.2-38.6) as calculated using the reverse censoring method of the product-limit estimate. The follow-up range was 5.6 to 123.5 months. At the time of data analysis, 87 patients (83.7%) were alive. Seventeen patients (16.3%) were deceased. Nine patients died of disease, while 8 died from another cause. At the time of last follow-up, 86 patients (82.7%) had no evidence of disease (Table 1).

Table 1. Recurrence, Disease Status, and Follow-up Data for 104 Patients (105 Tumors) With MCC Treated With Surgical Excision of the Primary Tumor Without Adjuvant RT

Characteristic	No. (%), Mean or Median
Follow-up, mo	
Median (range)	34.3 (5.6-123.5)
Status at last follow up	
No evidence of disease	86 (82.7)
Alive with disease	1 (1.0)
Died of disease	9 (8.7)
Died of other cause	8 (7.7)
Any recurrence	
Yes	19 (18.3)
No	85 (81.7)
Mean time to first recurrence, ^a mo	
Mean (range)	10.7 (3.5-23.3)
Site of recurrence	First/subsequent
Local	0/0
Satellite	0/0
In-transit	3 (2.9)/0
Regional	9 (8.7)/0
Distant	2 (1.9)/5 (4.8)
Local and in-transit	1 (1.0)/0
Satellite and regional	1 (1.0)/0
In-transit and regional	1 (1.0)/0
Regional and distant	2 (1.9)/1 (1.0)

^a Overall, 19 patients experienced recurrences.

Recurrences

Nineteen patients (18.3%) developed recurrent disease. The mean time to first recurrence was 10.7 months. True local recurrence was observed in 1 patient (1.0%). This patient also had concurrent in-transit recurrence. Satellite recurrence occurred in 1 patient (1.0%) who had concurrent regional recurrence. Four additional patients developed in-transit metastases as a component of their recurrence (3.8%).

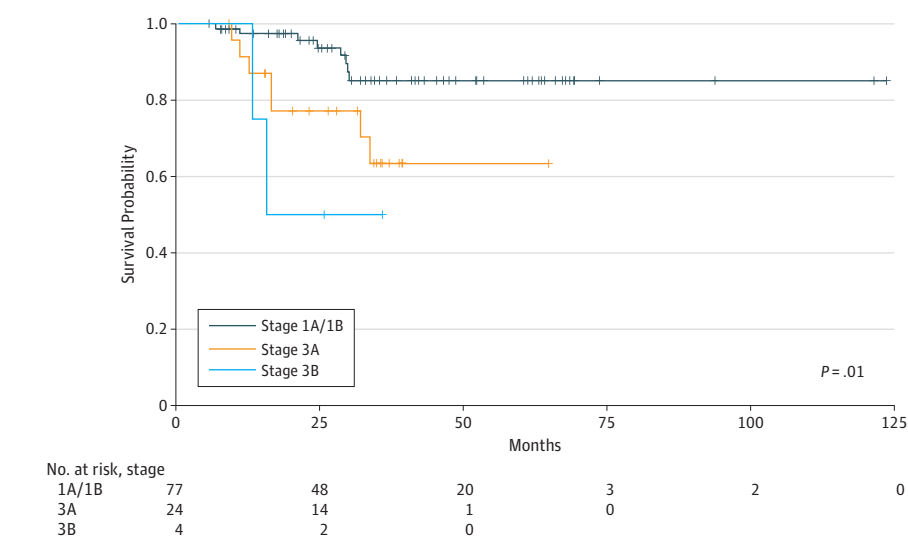
The regional nodal basin was the most common site of recurrence. Thirteen patients (12.5%) had a regional recurrence component. Distant metastases were observed in 4 patients (3.8%). Subsequent regional and/or distant recurrences developed in 6 patients. Recurrence data are summarized in Table 1.

Survival

When stratified by initial pathologic stage, the OS and MCCSS at 48 months were estimated to be 85.0% (95% CI, 71.8%-92.3%) and 94.4% (95% CI, 83.4%-98.2%) for patients with stage 1A/B disease and 63.2% (95% CI, 36.6%-81.1%) and 78.1% (95% CI, 50.0%-91.5%) for patients with stage 3A disease (Figure 1 and Figure 2). The OS and MCCSS at 24 months for patients with stage 3B disease were both 50.0% (95% CI, 5.8%-84.5%).

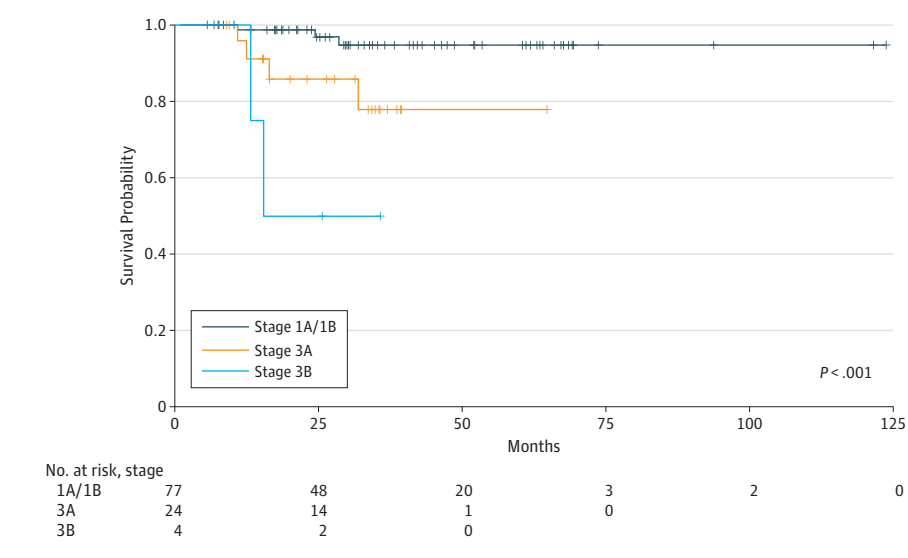
Overall survival was significantly associated with patient age at diagnosis and positive nodal disease (positive SLN or clinical nodal disease at presentation) (Table 2). Disease-free survival was significantly associated with positive nodal disease but not with patient age at diagnosis (Table 3). In the best

Figure 1. Overall Survival



This Kaplan-Meier curve shows overall survival stratified by pathologic stage of disease at presentation for 104 patients with Merkel cell carcinoma (105 tumors) treated with surgical excision of the primary tumor without adjuvant radiation therapy.

Figure 2. Merkel Cell Carcinoma-Specific Survival



This Kaplan-Meier curve shows Merkel cell carcinoma-specific survival stratified by pathologic stage of disease at presentation for 104 patients with Merkel cell carcinoma (105 tumors) treated with surgical excision of the primary tumor without adjuvant radiation therapy.

multivariate models, patients with nodal disease (stage 3A and 3B) were more than 4 times more likely to have disease failure and 3 times more likely to die than patients with node-negative disease (Tables 2 and 3).

Discussion

There is little debate that the current initial treatment of primary MCC typically includes surgical excision. Although surgical margins for primary MCC remain undefined, best practice margins are likely based on the size of the primary tumor and anatomic site. Historically, wide local excision with 2- to 3-cm margins was recommended.^{7,9,10} However, in the largest single-institution study³ that included the evaluation of lo-

cal recurrence, a relatively low local recurrence rate (8%) was achieved with an average margin-negative surgical excision of 1.1 cm. Boyer et al⁵ reported a marginal recurrence rate of 4%, defined as recurrent tumor contiguous with the primary excision scar, following Mohs surgery with a mean surgical margin of 1.67 cm for tumors averaging 1.58 cm in diameter.

The role of adjuvant RT to the primary site following surgical excision also remains undefined. The available literature is largely limited to retrospective single-institution studies and small case series with heterogeneous cohorts of patients. Several studies have shown statistically significant differences between local recurrence rates with surgery alone compared with surgery plus RT, favoring adjuvant RT.^{6,9,11-14} Yet others have demonstrated no difference in local recurrence rates between these groups.^{3-5,7,15} Disagreement regard-

Table 2. Characteristics Associated With Overall Survival for 104 Patients (105 Tumors) With MCC Treated With Surgical Excision of the Primary Tumor Without Adjuvant RT

Characteristic	HR (95% CI)	P Value
Univariate Associations		
Sex		
Male	2.31 (0.81-6.58)	.12
Female	1 [Reference]	
Age at diagnosis		
1-Year increase	1.06 (1.01-1.11)	.02
Body site		
Head/neck	1 [Reference]	.57
Lower extremity	0.65 (0.14-3.05)	
Upper extremity	1.45 (0.47-4.46)	
Trunk	0.50 (0.11-2.34)	
Diameter primary, cm ^a		
<1	1 [Reference]	.92
1-2	0.95 (0.35-2.57)	
Depth ^a		
1-mm increase	0.95 (0.78-1.15)	.59
Angiolymphatic invasion		
No	1 [Reference]	.92
Yes	0.94 (0.31-2.89)	
Mitoses		
1-Mitosis/mm ² increase	1.01 (0.98-1.02)	.67
Growth pattern		
Circumscribed	1 [Reference]	.78
Infiltrative	0.86 (0.29-2.52)	
Ulceration		
No	1 [Reference]	.31
Yes	2.14 (0.49-9.36)	
Positive nodal disease ^b		
No	1 [Reference]	.01
Yes	3.42 (1.32-8.86)	
Best Multivariate Model		
Age at diagnosis		
1-Year increase	1.05 (1.00-1.10)	.04
Positive nodal disease ^b		
No	1 [Reference]	.02
Yes	3.05 (1.17-7.97)	

Abbreviations: HR, hazard ratio; MCC, Merkel cell carcinoma; RT, radiation therapy.

^a Diameter and depth of the primary are highly correlated and were not offered simultaneously to the multivariate model.

^b Positive sentinel lymph node or clinical nodal disease at presentation.

ing the use of adjuvant RT to the primary site is not surprising. The variability of treatment and end points evaluated (any recurrence vs stratified local, regional, or distant recurrence vs survival) leads to difficulties in interpretation and comparison among published studies.

Several groups have studied the potential survival benefit from the use of adjuvant RT (local and/or regional) for patients with MCC. Some were able to demonstrate a positive correlation.^{12,14,16,17} Yet others were unable to detect

Table 3. Characteristics Associated With Disease-Free Survival for 104 Patients (105 Tumors) With MCC Treated With Surgical Excision of the Primary Tumor Without Adjuvant RT

Characteristic	HR (95% CI)	P Value
Univariate Associations		
Sex		
Male	2.08 (0.90-4.82)	.09
Female	1 [Reference]	
Age at diagnosis		
1-Year increase	1.03 (0.99-1.07)	.10
Body site		
Head/neck	1 [Reference]	.35
Lower extremity	0.69 (0.22-2.13)	
Upper extremity	1.12 (0.44-2.85)	
Trunk	0.29 (0.07-1.30)	
Diameter primary, cm ^a		
<1	1 [Reference]	.09
1-2	1.96 (0.89-4.30)	
Depth ^a		
1-mm increase	0.96 (0.82-1.18)	.58
Angiolymphatic invasion		
No	1 [Reference]	.24
Yes	1.65 (0.71-3.84)	
Mitoses		
1-Mitosis/mm ² increase	1.01 (0.99-1.02)	.19
Growth pattern		
Circumscribed	1 [Reference]	.32
Infiltrative	1.51 (0.67-3.43)	
Ulceration		
No	1 [Reference]	.19
Yes	2.24 (0.67-7.52)	
Positive nodal disease ^b		
No	1 [Reference]	<.001
Yes	4.27 (1.93-9.43)	
Best Multivariate Model		
Positive nodal disease ^b		
No	1 [Reference]	<.001
Yes	4.27 (1.93-9.43)	

Abbreviations: HR, hazard ratio; MCC, Merkel cell carcinoma; RT, radiation therapy.

^a Diameter and depth of the primary are highly correlated and were not offered simultaneously to the multivariate model.

^b Positive sentinel lymph node or clinical nodal disease at presentation.

an improved survival rate following the use of adjuvant RT.^{3,5-7,11,18-20} Again, confounding variables including surgical margins, sites radiated, additional treatments such as chemotherapy, and sites of recurrence compromise the validity of the conclusions. Mojica et al¹⁷ reported on 1665 cases of MCC in the SEER registry and concluded that the use of adjuvant RT is associated with improved survival in patients with MCC. However, the authors admit that multiple potential confounding variables such as chemotherapy use, surgical margin status, completion of adjuvant therapy, RT dose, sites radiated, and recurrence data were unable to

be evaluated. Moreover, after correcting for selection bias, adjuvant RT did not significantly improve OS.²¹

When considering the value of a particular therapy, the intent of the treatment intervention, whether local, regional, and/or distant disease control, should drive the evaluation of its effectiveness. At our institution, the use of adjuvant RT to the primary tumor site is based on the potential improvement of local control. In our opinion, the use of adjuvant RT to the primary site to prevent regional or distant recurrence and improve survival in the absence of local recurrence is counterintuitive. Given the paucity of high-level evidence to guide MCC primary site management, we developed an institutional algorithmic approach to the management of patients with AJCC stage 1 to 3 MCC, including management of the primary site.²² For tumors less than 2 cm in clinical diameter, wide local excision with 1- to 2-cm margins is recommended without adjuvant RT to the primary site, thereby sparing any associated morbidity. For tumors 2 cm or larger, wide excision with 2-cm margins is recommended when possible, followed by adjuvant RT to improve local control. These surgical margins are generally attainable without significant cosmetic or functional morbidity (eFigures 1 and 2 in the [Supplement](#)). All positive margins are re-excised if possible to achieve negative histopathologic margins. Other indications at our institution for adjuvant RT to the primary tumor site may include, in some cases, the presence of other adverse features like angiolymphatic invasion and/or a close surgical margin where re-excision is not feasible. Every patient is discussed at our Multidisciplinary MCC Tumor Board to determine individualized optimal treatment. Mohs surgery is generally not performed for MCC at our institution in part owing to the logistical difficulty of coordinating with SLN biopsy, which is performed under general anesthesia. Moreover, excellent local control and final cosmetic outcomes can be achieved without Mohs surgery (eFigures 1 and 2 in the [Supplement](#)).

The primary aim of this study was to evaluate the occurrence of local recurrence (including true local and satellite recurrence) following wide local excision of the primary tumor without the use of adjuvant RT in a contemporary cohort of patients with MCC. The patients in our study had primary tumors less than 2 cm in diameter, consistent with our management approach. True local recurrence was uncommon (1 of 104 patients [1.0%]). When satellite recurrences were included, local recurrence occurred in only 2 patients (1.9%). These recurrences would have been within a standard radiation field with a 5-cm radius and therefore could potentially have been prevented with the use of adjuvant RT to the primary site. In-transit recurrence was observed in 5 patients (4.8%). The potential benefit of adjuvant RT to the primary tumor site in patients who later develop in-transit recurrences in the absence of local recurrence is questionable, unless the in-transit disease falls exclusively within the 5-cm radius standard radiation field. Unfortunately, in-transit disease is often unpredictable and can occur many centimeters from the primary site. Four of the 5 patients with in-transit recurrences had positive SLNs at the time of treatment of the primary tumor. Two of these patients with positive SLNs had in-transit disease falling outside a 5-cm radius around the primary lesion

site. One patient had a single in-transit metastasis falling within a 5-cm radius. However, while receiving adjuvant RT to the primary and in-transit sites, the patient developed additional in-transit metastases outside the radiation field. For 1 patient, accurate location of the in-transit disease could not be determined. For the patient who was SLN negative with an in-transit recurrence, disease occurred beyond a 5-cm radius and was accompanied by palpable regional disease. Unless the radiation field would have included the primary tumor bed, the entire lymphatic drainage area, and the nodal basin (which off the head and neck is generally not achievable), adjuvant RT to the primary site would not be expected to have prevented the in-transit recurrences in at least 3 of 5 of these patients.

Of the 19 patients with recurrences, 12 (63%) presented with AJCC stage 3A or 3B disease. Three patients had palpable nodal disease on presentation (3B) and 9 patients had nodal metastases identified on SLN biopsy (3A). For these 12 patients with known regional metastatic disease, it would be difficult to rationalize that adjuvant RT to the primary site would offer any additional benefit beyond local control. The other 7 patients (those without nodal disease at presentation) recurred regionally or distantly, none recurred locally. In the absence of a local or satellite recurrence, it would again be difficult to argue that adjuvant RT to the primary site would have decreased the risk of regional recurrence or distant metastasis.

Stratified by pathologic stage, the 4-year MCCSSs were 94% for stage 1A/B ($n = 77$) and 78% for stage 3A ($n = 24$). The 2-year MCCSS for stage 3B was 50% ($n = 4$). Based on multivariate analysis, DFS was only associated with nodal disease at presentation (either clinical nodal disease or a positive SLN), while OS was associated with both nodal disease at presentation and patient age at diagnosis.

Conclusions

Merkel cell carcinoma is a radiosensitive tumor, and adjuvant RT plays a critical role in the management of high-risk local disease and/or regional disease. In the setting of a high-volume MCC clinic with an experienced multidisciplinary team, we have found a low occurrence of true local and satellite recurrences (2 of 104 patients [1.9%]) in selected patients, with primary tumors less than 2 cm in diameter treated with surgery alone without adjuvant RT to the primary site. This is an observational study of patients with low-risk small primary MCC tumors who did not receive adjuvant RT to the primary site in their initial management. Based on our treatment algorithm, there is no control group with similar MCC tumors managed with surgery and adjuvant RT as initial treatment of the primary lesion. However, the relative low occurrence of local recurrence in our cohort suggests that adjuvant RT to the primary site may be unnecessary in patients with low-risk small primary lesions. When a paucity of high-level evidence exists, critical review of the best available evidence is imperative before recommending any type of treatment, surgical or nonsurgical, particularly when those treatments have potential morbidity and significant cost.

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Acquisition, analysis, or interpretation of data: Frohm, Griffith, Harms, Hayman, Fullen, Nelson, Wong, Schwartz, Bichakjian.

Drafting of the manuscript: Frohm, Griffith, Schwartz.

Critical revision of the manuscript for important intellectual content: Frohm, Griffith, Harms, Hayman, Fullen, Nelson, Wong, Schwartz, Bichakjian.

Statistical analysis: Griffith, Harms.

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